

Lecture 12: Survival Models and Insurance Pricing

Phase 3: Actuarial Science and Insurance Principles

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In Lecture 11, we valued payments using time, discounting, and life-table probabilities. Today we ask where survival probabilities come from and how insurers combine survival, claims, and loading into a premium.

From Lecture 11 to Lecture 12 I

- ▶ Lecture 11 built a **present-value engine**: move a future cash flow back to today.
- ▶ Lecture 11 also built a **life-table engine**: calculate survival and death probabilities from survivor counts.
- ▶ Lecture 12 connects these engines to three modelling questions:
 1. How do we model **time until an event**?
 2. How do we model **how many claims** occur?
 3. How do we model **how large claims** are?

Key Definition / Result

The big idea is not a single formula. It is a system: **timeline + probability model + claim model + pricing decision**.

Opening problem: survival, claims, and premiums I

Cognitive Trigger / Opening Problem

A school is considering a simple protection plan for tablets used in class. Some years there may be no claims, some years there may be a few cracked screens, and in some years a larger accident may create many claims.

At the same time, for a life-related product, whether a payment is made depends on who is still alive at a future date.

Questions:

1. How can we describe the chance that an event has not happened yet?
2. How can we model *how many* claims occur and *how large* the claims are?
3. How can all of these pieces be combined into a premium that is sensible and explainable?

Opening problem: survival, claims, and premiums II

Proof / Modelling Decision Point

Choice. Do we model the world by *one giant formula*, or by *separate moving parts* such as time, count, size, probability, and discounting?

Why this choice matters. A modular model is easier to debug, interpret, and improve. Strong actuarial thinking means understanding the architecture, not only the arithmetic.

Why this topic matters I

- ▶ Survival models turn a vague question, “how long until something happens?”, into a structured mathematical object.
- ▶ Insurance pricing combines time, probability, claim behaviour, expenses, and risk judgement.
- ▶ Claims data contain two different stories: **frequency** and **severity**.
- ▶ Technology lets us run simulations and see patterns instead of memorising heavy formulas.

Why this topic matters II

Key Definition / Result

Big theme of the day: a premium is never just one number. It is the result of assumptions, data, timing, uncertainty, and interpretation.

Pause-and-Think

If two insurers use the same data but choose different models, can both still sound reasonable? What does that tell us about mathematics as a decision tool?

Roadmap for Lecture 12 I

1. From life tables to survival curves.
2. Survival functions, hazard intuition, and discrete-time survival.
3. Life expectancy revisited through survival thinking.
4. Claims frequency: how many claims?
5. Claims severity: how large are claims?
6. Aggregate loss and premium pricing.
7. Technology, sanity checks, and investigations.

Key Definition / Result

The order matters: first we model whether and when events occur, then we model claim counts and sizes, and finally we price the risk.

The pricing spine in one sentence I

Key Definition / Result

Pricing idea:

price = present value of possible payments $\times$ probability pattern behind those payments.

- ▶ For an annuity, payments depend on being alive on future dates.
- ▶ For non-life insurance, annual loss depends on how many claims occur and how large those claims are.
- ▶ We will use **tables, timelines, and simulations**, not calculus derivations.

Pause-and-Think

Is the sentence above literally one multiplication in every situation, or is it a conceptual reminder to combine time and uncertainty carefully?

Three lenses for understanding risk products I

Key Definition / Result:
Timing

When can money move?

Investigation / Activity:
Chance

Will the payment event occur?

Working Example: Size

If it occurs, how large is it?

- ▶ Survival models sharpen the **chance** and **timing** lenses.
- ▶ Claim models sharpen the **how many** and **how large** lenses.
- ▶ Premium design forces all lenses to speak to each other.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Explain the three-lens idea of timing, chance, and size using a metaphor from music, where a payment is like a sound event. Do not use calculus. Give one insurance example and one everyday-life example.

Part A. From Life Tables to Survival Curves



Recall: what did the life table give us? I

- ▶ Lecture 11 used l_x , d_x , p_x , and q_x to describe survival and death year by year.
- ▶ A life table is already a survival model in table form.
- ▶ Today we turn that table into a curve and then into more general modelling language.

Key Definition / Result

Life-table idea:

$${}_n p_x = \frac{l_{x+n}}{l_x}.$$

This is already a survival probability from age x to age $x + n$.

From life table to survival curve I

- ▶ If we divide each future survivor count by the starting count, we get a survival proportion.
- ▶ Plotting these survival proportions against age or time gives a survival curve.
- ▶ The curve is a visual version of the life table.

Technology Check

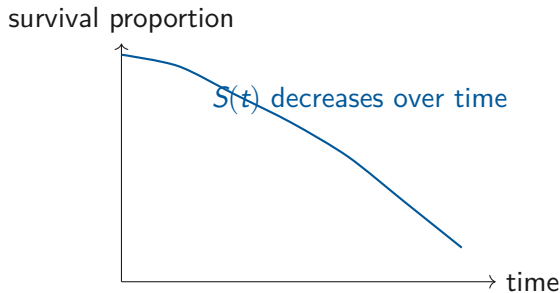
In Excel or Google Sheets, if the starting survivor count is in cell B2, a survival proportion column can use a formula such as `=B2/$B$2`, then be filled downward.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

I have a life table with age and survivor count l_x . Explain, in plain language for gifted secondary students, why dividing each l_x by the starting count creates a survival curve. Then suggest two questions that compare the shape of two such curves.

What a survival curve usually looks like I



- ▶ Survival starts high and tends to fall as time increases.
- ▶ A steeper drop means the event is arriving more quickly.
- ▶ If a survival curve increases, something in the data, formula, or interpretation should be checked.

What is a survival model? I

- ▶ A survival model tries to describe **time until an event happens**.
- ▶ The event could be death, failure of a device, first insurance claim, or time until a benefit starts.
- ▶ It is a model of waiting time.

Key Definition / Result

Typical actuarial question:

What is the chance that the event has not happened by time t ?

Pause-and-Think

What changes if the event is not “death” but “first major claim”, “device failure”, or “first missed payment”? Which parts of the mathematics stay the same?

A metaphor: a candle and a stopwatch I

- ▶ Imagine each life or device as a candle.
- ▶ The stopwatch records time; the candle tells us whether the event has happened.
- ▶ A survival model describes how many candles are still burning as time passes.

Key Definition / Result

A survival curve is like the **outline of remaining candles over time**.

Proof / Modelling Decision Point

Choice. Should we pay attention to *individual candles* or to the *whole candle population*?

Why this choice matters. Actuarial pricing usually works at the portfolio level, but understanding the portfolio starts from a good individual waiting-time model.

Part B. Survival Functions, Hazard, and Discrete-Time Survival

The survival function I

Key Definition / Result

If T is the time until an event, then the survival function is

$$S(t) = P(T > t).$$

- ▶ Read it as: the probability that the event time is **later than** t .
- ▶ In other words, by time t , the event has **not yet happened**.
- ▶ We can talk about the idea clearly even if we do not study continuous-time calculus.

The survival function II

Proof / Modelling Decision Point

Choice. Why write $P(T > t)$ instead of immediately jumping to “probability of death by time t ”?

Why this choice matters. Survival language is often more natural for annuities, equipment reliability, and waiting-time interpretation. It encourages curve thinking instead of single-point thinking.

Worked Example 1: interpreting a survival value I

Worked Example

Suppose $S(2) = 0.92$.

1. The symbol $S(2)$ refers to the survival function at time 2.
2. The number 0.92 means a probability of 92%.
3. So the event has not happened by time 2 with probability 0.92.

Interpretation: there is a **92%** chance that the life or device survives beyond time 2.

Working Example

If $S(2) = 0.92$, the statement “survival beyond time 2 is very likely” is sensible.

Boundary / Failing Example

If someone says “ $S(2) = 1.08$ because the curve starts near 1 and looks high”, the statement is impossible: probabilities cannot exceed 1.

Worked Example 1: interpreting a survival value II

Investigation / Activity: Boundary Condition

- ▶ **Critical condition highlighted.** A survival value must stay between 0 and 1.
- ▶ **Thinking direction.** Many modelling errors are hidden violations of basic probability rules.

Hazard in plain language I

- ▶ Hazard means the local or stage-by-stage risk that the event happens around a given time.
- ▶ In a school-friendly version, we can think of hazard as “how risky this stage is”.
- ▶ No calculus formula is needed for the main intuition.

Key Definition / Result

Plain-language hazard:

the chance of the event arriving soon, given that it has not happened yet.

Proof / Modelling Decision Point

Choice. Why does the phrase “given that it has not happened yet” matter?

Why this choice matters. Because risk at a stage should only be judged among those still exposed to that stage. Hazard is a conditional idea, not an unconditional one.

Discrete-time survival: year by year I

- ▶ For our setting, a year-by-year model is very natural.
- ▶ In each year, there are only two outcomes:
 1. survive the year;
 2. die, fail, or experience the event during the year.
- ▶ This is a **Bernoulli-style step** repeated across years.

Pause-and-Think

If the risk changes with age, do we still keep the year-by-year structure? Yes, but the yearly probabilities may no longer be identical. The structure can survive even if a simple shortcut does not.

Bernoulli thinking for one year I

Key Definition / Result

For one year:

$$P(\text{survive}) = p, \quad P(\text{event}) = q, \quad p + q = 1.$$

- ▶ This is the same yes/no logic students have seen in Bernoulli examples.
- ▶ Survival modelling can therefore start in a familiar place.

Working Example

A yearly tablet-break model with outcomes “claim” or “no claim” in one short period fits Bernoulli logic.

Boundary / Failing Example

A yearly total claim amount in dollars does *not* fit Bernoulli logic because it is not a yes/no variable.

Bernoulli thinking for one year II

Investigation / Activity: Boundary Condition

- ▶ **Critical condition highlighted.** Bernoulli models answer one binary question only.
- ▶ **Thinking direction.** Good modelling begins by matching the question type to the variable type.

If yearly risk is constant, a geometric model appears!

Key Definition / Result

If the annual event probability is always the same value q , then the number of years until the first event can be modelled by a **Geometric distribution**.

- ▶ “First success” here means the first year in which death, failure, or the target event occurs.
- ▶ The model is simple because every year asks the same probability question again.

Proof / Modelling Decision Point

Choice. What major assumption creates the geometric model?

Why this choice matters. The annual event probability must stay constant from year to year. If risk changes with age or wear, the geometric shortcut may stop being appropriate even though year-by-year thinking still works.

A metaphor for the geometric model I

- ▶ Picture a row of identical locked doors, one door for each year.
- ▶ Each year, the same small chance q opens the event door.
- ▶ The geometric model records **which year the first event door opens**.

Key Definition / Result

The model is memoryless in plain language: every year asks the **same probability question again**.

Pause-and-Think

Is “no memory” realistic for human lives? For machine parts? For tablet screens? Where does this model sound elegant, and where does it sound suspicious?

Worked Example 2: geometric survival in words I

Worked Example

Suppose the annual death probability is $q = 0.02$ and is assumed constant.

1. One-year survival probability is $p = 0.98$.
2. Surviving 3 full years means surviving year 1, year 2, and year 3.
3. Under the constant-yearly-risk model,

$$P(\text{survive at least 3 years}) = 0.98^3.$$

4. Numerically this is about **0.9412**.

Interpretation: under this simplified model, survival beyond 3 years has probability about 94.12%.

Working Example

Using 0.98^3 is valid when the yearly survival chance remains 0.98 each year.

Boundary / Failing Example

If the yearly survival chances are 0.99, 0.98, 0.96, then 0.98^3 is wrong; the correct product is $0.99 \times 0.98 \times 0.96$.

Worked Example 2: geometric survival in words II

Investigation / Activity: Boundary Condition

- ▶ **Critical condition highlighted.** The geometric shortcut only works under constant-yearly risk.
- ▶ **Thinking direction.** A beautiful formula can fail quietly when one hidden condition fails.

Optional extension: exponential as a continuous-time cousin I

- ▶ If time is treated as continuous rather than year by year, a common model is the [Exponential distribution](#).
- ▶ We use it only as a conceptual extension. No derivation is needed.
- ▶ It plays a role similar to the geometric model, but in continuous time.

Key Definition / Result

Keep the message simple: geometric is the [discrete-time memoryless model](#); exponential is its [continuous-time analogue](#).

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Explain the relationship between the geometric distribution and the exponential distribution using a transport metaphor. Focus on waiting time, memorylessness, and discrete versus continuous time. Do not derive formulas.

Why model choice matters I

- ▶ If survival declines faster, annuities become less valuable.
- ▶ If deaths or failures are expected earlier, some benefits become more likely and some become less likely.
- ▶ A poor survival model can misprice products.

Key Definition / Result

This is an early taste of **model risk**: if the model is wrong, the premium can be wrong too.

Pause-and-Think

Which is more dangerous: a calculation mistake inside a good model, or a perfect calculation inside a bad model? Defend your answer.

Common mistake: survival function versus point-event intuition I

Common Mistake / Pitfall

Students sometimes confuse:

- ▶ the chance of surviving *beyond* a time;
- ▶ the chance that the event happens *at* a specific time;
- ▶ the general shape of a survival curve.

Keep returning to one question:

Has the event happened by time t or not?

Working Example

“ $S(4) = 0.81$ means 81% survive beyond time 4” is correct.

Boundary / Failing Example

“ $S(4) = 0.81$ means the event happens exactly at time 4 with probability 81%” is incorrect.

Common mistake: survival function versus point-event intuition II

Investigation / Activity: Boundary Condition

- ▶ **Critical condition highlighted.** Survival $\neq$ point-event probability.
- ▶ **Thinking direction.** When a concept is conditional on time, every phrase in the sentence matters.

Part C. Life Expectancy Revisited Through Survival Thinking



Life expectancy revisited I

- ▶ Lecture 11 introduced life expectancy using survivor counts.
- ▶ Survival modelling gives a broader way to think about the same idea: expected lifetime depends on the entire survival pattern, not only one age interval.
- ▶ If survival remains high for longer, life expectancy is larger. If survival drops quickly, life expectancy is smaller.

Key Definition / Result

Life expectancy is a summary of the whole survival story.

Backward recursion as survival logic I

Key Definition / Result

Recall the student-friendly recursion:

$$e_x = p_x(1 + e_{x+1}).$$

- ▶ To enjoy one more complete year and then the future expectation after that, the life must survive the next year.
- ▶ This makes life expectancy a conditional-survival idea.
- ▶ The same logic appears again in life annuities and insurance pricing.

Pause-and-Think

If p_x becomes smaller at older ages, how should this affect life expectancy? Explain before calculating.

Part D. Claims Frequency: How Many Claims?

Claims frequency asks “How many?” I

- ▶ Claims frequency counts the number of claims in a period.
- ▶ It is a counting question, not a cost question.
- ▶ Examples include the number of cracked screens this month, the number of medical claims this year, or the number of accidents in a quarter.

Pause-and-Think

Which data column answers “how many” and which answers “how costly”? If your workbook mixes them in one column, the modelling mistake starts before any formula appears.

Frequency models students can recognise I

Key Definition / Result:
Bernoulli

Claim or no claim.

Investigation / Activity: Bi-
nomial

How many claims in a fixed number of independent tries.

Working Example: Poisson

How many events in a time window.

Proof / Modelling Decision Point

Choice. Which model should be chosen first: Binomial or Poisson?

Why this choice matters. Ask what the story gives you. A fixed number of trials suggests Binomial; a time window with random arrivals often suggests Poisson.

When Poisson becomes a useful classroom model I

- ▶ If claims arise over time and we want a simple count model, Poisson is a classic starting point.
- ▶ It is popular because one parameter already says a lot about the average count level.
- ▶ Students may already have seen it in earlier probability-distribution lessons.

Key Definition / Result

If $N \sim \text{Poisson}(\lambda)$, then λ is the average number of claims in the period.

Pause-and-Think

Why might a one-parameter model still be powerful? Sometimes a simple model is not weak; it is a disciplined first approximation.

Worked Example 3: interpreting a Poisson mean I

Worked Example

Suppose an insurer expects on average $\lambda = 2$ claims in a month.

1. The symbol λ is the expected monthly count.
2. So the model says the monthly claim count centres around 2.
3. The month-to-month number may vary, but 2 is the average level.

Interpretation: the portfolio is not promising exactly 2 claims each month; it is promising an **average rate of about 2**.

Working Example

“ $\lambda = 2$ ” means a typical month is organised around an average of 2 claims.

Boundary / Failing Example

“ $\lambda = 2$ ” means every month must contain exactly 2 claims is false.

Worked Example 3: interpreting a Poisson mean II

Investigation / Activity: Boundary Condition

- ▶ **Critical condition highlighted.** An expected value is a centre, not a guarantee.
- ▶ **Thinking direction.** The word *average* is one of the most abused words in applied mathematics.

Worked Example 4: Poisson probabilities I

Worked Example

Let $N \sim \text{Poisson}(2)$ for the monthly number of claims.

1. Probability of no claims:

$$P(N = 0) = e^{-2} \approx 0.1353.$$

2. Probability of exactly 3 claims:

$$P(N = 3) = e^{-2} \frac{2^3}{3!} \approx 0.1804.$$

3. We do not derive this formula; technology can evaluate it quickly.

Interpretation: under this model, 0 claims is possible, and 3 claims is also a realistic monthly outcome.

Worked Example 4: Poisson probabilities II

Proof / Modelling Decision Point

Choice. Why calculate both $P(N = 0)$ and $P(N = 3)$?

Why this choice matters. Because one probability near the left edge and one from the middle begin to reveal the shape of the distribution, not just a single answer.

Technology check: Poisson in spreadsheet or R I

Technology Check: Excel Examples

- ▶ `=POISSON.DIST(0,2,FALSE)`
- ▶ `=POISSON.DIST(3,2,FALSE)`

Technology Check: Optional R

```
1 dpois(0, lambda = 2)
2 dpois(3, lambda = 2)
```

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Explain, for gifted secondary students, why a Poisson model can make sense for claim counts in a time period. Then provide an Excel mini-table that calculates $P(N=k)$ for $k=0,1,2,3,4,5$ when $\lambda=2$. Do not derive the formula.

Part E. Claims Severity: How Large Are Claims?

Claims severity asks “How large?” I

- ▶ Frequency counts *how many* claims happen.
- ▶ Severity measures *how expensive* each claim is.
- ▶ The two questions are related but different.

Key Definition / Result

A portfolio can have low frequency but high severity, or high frequency but low severity.

Pause-and-Think

Which sounds more dangerous to an insurer: many tiny claims or one huge claim? Why is the answer not purely mathematical?

A practical school-level severity model I

- ▶ We start with discrete severity levels with probabilities.
- ▶ This is transparent and easy to simulate.
- ▶ Example severity levels: HK\$500, HK\$1,500, HK\$4,000.

Key Definition / Result

A discrete severity table is often the best first classroom model because students can inspect every row.

Proof / Modelling Decision Point

Choice. Why use a discrete table before introducing sophisticated continuous severity models?

Why this choice matters. Because interpretation comes first. When students can inspect every row, they can detect hidden assumptions and impossible probabilities more easily.

Worked Example 5: expected claim size from a table

Worked Example

Suppose claim size X has:

Size	Probability	Contribution
500	0.60	500×0.60
1500	0.30	1500×0.30
4000	0.10	4000×0.10

1. Multiply each size by its probability.
2. Add the contributions.
3. The expected severity is

$$E(X) = 500(0.60) + 1500(0.30) + 4000(0.10) = \mathbf{1150}.$$

Interpretation: HK\$1,150 is the average claim size under this simplified model.

Why severity can be skewed I

- ▶ Many small claims can occur, but a few very large claims can also appear.
- ▶ This creates a distribution with a long right tail.
- ▶ We can discuss the shape without advanced distribution theory.

Key Definition / Result

Plain message: **insurance losses are often not symmetric**. Large rare claims matter.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Explain skewed insurance claim severity using a city-traffic metaphor. Then create one classroom example where the mean is pulled upward by a rare large loss. Use simple numbers only.

Part F. Aggregate Loss and Premium Pricing

Aggregate loss combines count and size I

- ▶ If N is the number of claims and $X_1, X_2, \dots$ are claim sizes, then the total annual loss is the sum of all claim amounts in that year.
- ▶ This is the natural bridge from individual claims to portfolio pricing.

Key Definition / Result

Aggregate loss idea:

$$L = X_1 + X_2 + \dots + X_N.$$

If there are no claims, the sum is zero.

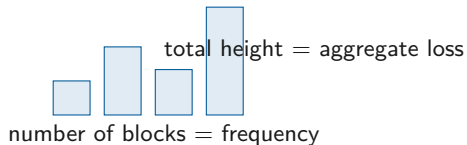
Proof / Modelling Decision Point

Choice. Why put frequency and severity into a *single* total-loss variable?

Why this choice matters. Because pricing is usually done for the total cost borne by the insurer, not separately for counts and individual claim sizes.

A visual metaphor for aggregate loss I

- ▶ Think of annual loss as a stack of blocks.
- ▶ The **number of blocks** comes from frequency.
- ▶ The **height of each block** comes from severity.
- ▶ Total stack height is aggregate loss.



Pause-and-Think

If the number of blocks stays the same but one block becomes huge, which part of the model failed to warn us: frequency, severity, or both?

Expected loss as the first pricing anchor I

Key Definition / Result

A very useful school-level approximation is

$$E(\text{annual loss}) \approx E(N) \times E(X).$$

- ▶ This gives a first benchmark for pricing.
- ▶ Later a loading or safety margin can be added.
- ▶ It connects frequency, severity, and premium in one clear line.

Proof / Modelling Decision Point

Choice. Why call this a *first anchor* rather than the final answer?

Why this choice matters. Because expected loss captures the centre of the distribution, not the full uncertainty, business expense, or caution level.

Worked Example 6: expected annual loss I

Worked Example

Suppose the expected number of claims per year is $E(N) = 0.20$ and the expected claim size is $E(X) = 6000$.

1. Use the benchmark formula:

$$E(\text{annual loss}) \approx E(N)E(X).$$

2. Substitute the values:

$$0.20 \times 6000 = \mathbf{1200}.$$

Interpretation: the expected annual claim cost per policy is about HK\$1,200.

Working Example

The quantity 0.20 is an expected count per policy-year, so multiplying by dollars per claim gives dollars per policy-year.

Boundary / Failing Example

If a student multiplies two dollar amounts, the units become nonsense for expected annual loss.

Worked Example 6: expected annual loss II

Investigation / Activity: Boundary Condition

- ▶ **Critical condition highlighted.** Units are part of the mathematics.
- ▶ **Thinking direction.** One of the fastest ways to catch an actuarial mistake is to ask: what are the units?

Premiums need more than expected loss I

- ▶ If an insurer charges only the expected loss, there may be no cushion for volatility, administration, or uncertainty.
- ▶ A premium often includes expenses and a loading.
- ▶ This is a business and risk-management decision, not just a probability calculation.

Key Definition / Result

Simple pricing formula:

$$\text{Premium} = \text{Expected Loss} + \text{Expenses} + \text{Loading}.$$

Pause-and-Think

If two insurers face the same expected loss but have different risk appetites, should they charge the same premium? Why or why not?

Worked Example 7: premium with expense and loading I

Worked Example

Suppose expected loss is HK\$1,200, fixed expense is HK\$150, and loading is 20% of expected loss.

1. Loading amount:

$$0.20 \times 1200 = 240.$$

2. Add all pieces together:

$$\text{Premium} = 1200 + 150 + 240.$$

3. Therefore,

$$\text{Premium} = \mathbf{1590}.$$

Interpretation: HK\$1,590 is a basic gross premium under these assumptions.

A more conservative simulated premium idea I

- ▶ One premium choice is the mean loss.
- ▶ Another is the mean loss multiplied by $(1 + \text{loading})$.
- ▶ A more conservative discussion can use a high percentile from simulation.

Key Definition / Result

This helps students see that pricing can depend on **how cautious** the insurer wants to be.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Act as an actuary explaining to gifted secondary students why an insurer might choose a premium above the mean simulated annual loss. Compare using the mean, a loaded mean, and a high percentile. Use plain language, not calculus.

Worked Example 8: life-contingent pricing meets insurance pricing I

Worked Example

Suppose a benefit of HK\$10,000 is paid at the end of year 2 if a person survives 2 years and then a claim event occurs. Let $v^2 = 0.9070$, let survival to time 2 be 0.96, and let the event probability at that payment point be 0.04.

1. Start with the potential payment amount: 10,000.
2. Apply the discount factor: multiply by 0.9070.
3. Apply the probability pattern: multiply by 0.96×0.04 .
4. Value:

$$10000 \times 0.9070 \times 0.96 \times 0.04 \approx \mathbf{348.29}.$$

Interpretation: a large nominal payment can have a much smaller present value when timing and event conditions both matter.

Worked Example 8: life-contingent pricing meets insurance pricing II

Proof / Modelling Decision Point

Choice. Why multiply several factors instead of subtracting from the nominal benefit?

Why this choice matters. Because discounting and probability each scale the payment by a proportion. This frame is the meeting point of life-contingent PV logic and event-conditional pricing logic.

Part G. Sanity Checks and Investigation

Sanity checks every model should pass I

Key Definition / Result

Useful automated checks:

1. Probabilities sum to 1.
2. Survival proportions do not increase over time.
3. Premium increases when loading increases.
4. Units are consistent: counts versus HK\$ amounts.

Proof / Modelling Decision Point

Choice. Why teach sanity checks so explicitly?

Why this choice matters. Because errors can survive inside technically correct formulas when the workbook architecture has not been challenged. Sanity checks convert scepticism into a habit.

Common mistake: mixing frequency and severity I

Common Mistake / Pitfall

Students often say “the average claim is 3” when they really mean 3 *claims*, not 3 dollars.

- ▶ Frequency is a **count**.
- ▶ Severity is an **amount**.
- ▶ Aggregate loss mixes both, but they should not be confused.

Working Example

“Average number of claims = 3” is a frequency statement.

Boundary / Failing Example

“Average claim = 3” is incomplete unless the unit is an amount or the sentence explicitly says “3 claims”.

Common mistake: mixing frequency and severity II

Investigation / Activity: Boundary Condition

- ▶ **Critical condition highlighted.** Mathematical language must carry units and object type.
- ▶ **Thinking direction.** Ambiguous language is often the first sign of ambiguous thinking.

Common mistake: over-trusting a single simulated run I

Common Mistake / Pitfall

A single simulation run may be noisy.

- ▶ One run is a story.
- ▶ Many runs reveal a pattern.
- ▶ This is where the Law of Large Numbers and Central Limit Theorem become conceptually relevant.

Pause-and-Think

Why do humans trust vivid single examples too easily? In quantitative work, which is more persuasive: one dramatic simulation, or a stable pattern across many runs?

LLN and CLT reappear in an actuarial setting I

- ▶ With many policies, average loss becomes more stable. This is the Law of Large Numbers intuition.
- ▶ The total or average loss becomes easier to describe statistically. This is where the Central Limit Theorem becomes conceptually useful.
- ▶ We do not need proofs today; simulation will make the idea visible.

Key Definition / Result

Why insurers like scale: **large portfolios make averages steadier and pricing more defensible.**

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Explain, without proofs, why LLN and CLT matter for insurers pricing a large portfolio. Use one metaphor from sport or crowd behaviour and one from quality control in manufacturing.

Investigation 1: compare two portfolios I

Investigation / Activity

Portfolio A: low frequency, high severity.

Portfolio B: high frequency, low severity.

1. Which portfolio feels more predictable?
2. Which one might justify a larger loading?
3. Build a toy simulation for both and compare the histogram of annual loss.

Pause-and-Think

Can two portfolios have the same expected loss but very different *stress level* for the insurer?
If yes, expected loss alone is not the whole story.

Investigation 2: match a model to a story I

Investigation / Activity

Match the situation to a model:

1. Years until an event, constant yearly risk.
2. Number of claims in a month.
3. Claim or no claim in one short period.
4. Cost level of a claim from a simple payout table.

Candidates: Bernoulli, Geometric, Poisson, discrete severity table.

Proof / Modelling Decision Point

Choice. Why is model-matching an important mathematical skill?

Why this choice matters. Because formulas do not announce themselves. Real modelling starts from recognising the hidden structure of the story.

Investigation 3: survival curve detective work I

Investigation / Activity

Provide two survival curves with different shapes.

1. Which curve represents faster event arrival?
2. Which one would give a lower value to a life annuity?
3. Which one would make an early-death insurance benefit more likely?

Explain each answer in words before using formulas or software.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Give me two contrasting survival-curve stories in words only: one with faster early decline and one with slower early decline. Then ask me three questions about annuity value, death-benefit likelihood, and model choice, without giving the answers.

Review Set 12A: concepts I

1. What does $S(t)$ mean?
2. Explain the word “hazard” in plain language.
3. How does a life table become a survival curve?
4. What is the difference between claim frequency and claim severity?
5. Why can a survival model affect annuity prices?
6. What does a loading do in a premium?
7. Why should probabilities and units be checked automatically?

Investigation / Activity: Depth Challenge

After answering each item, add one sentence beginning with: *“This would fail if ...”*

Review Set 12A: short computations I

8. If $S(4) = 0.81$, interpret it.
9. If yearly survival probability is 0.97 and constant, find survival beyond 2 years.
10. If $E(N) = 0.3$ and $E(X) = 2000$, estimate expected annual loss.
11. If expected loss is 600, expense is 80, and loading is 10% of expected loss, find the premium.
12. Use a severity table to compute an expected claim size from three rows of your own design.

Pause-and-Think

After each computation, ask: Which assumption did I use? Which unit did I produce? If you cannot answer both, the computation is incomplete.

Review Set 12B: richer reasoning I

1. Explain why a life annuity and a non-life insurance premium both need probability and time.
2. Why is the geometric model a natural discrete-time survival model for S2–S3 learners?
3. What could go wrong if frequency is modelled well but severity is modelled badly?
4. Why might an insurer prefer a premium above the mean simulated loss?

Investigation / Activity: Reasoning Extension

For each question, build a two-part answer:

1. the direct answer;
2. one counterexample or limiting case that tests the boundaries of your answer.

Review Set 12B: challenge tasks I

1. Design a spreadsheet with columns that separate count, size, and annual total loss clearly.
2. Sketch how you would compare a life-table survival curve with a simulated geometric survival curve.
3. Write a one-sentence actuarial explanation of *model risk* for gifted secondary students.
4. Design one sanity check that would detect an impossible probability table in your pricing workbook.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Act as a mathematical mentor. For each of the four challenge tasks, suggest two different approaches, without fully solving them. Emphasise where a student must make a modelling decision rather than just compute.

Key Definition / Result

Today we connected five ideas:

1. life tables can become **survival curves**;
2. survival models tell us **when events may occur**;
3. frequency models tell us **how many claims may occur**;
4. severity models tell us **how large claims may be**;
5. pricing combines them into a **premium decision**.

► You should leave with a system, not just isolated formulas.

► The system is:

timeline + survival model + claim model + simulation + interpretation.